

Statistical Analysis and Exploratory Data Analysis of US Healthcare Data

A Business Intelligence Report Using Python



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Tools Used: **Pandas | NumPy | Matplotlib | Seaborn | SciPy**

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Table of Contents

- Introduction
- Business Objective
- Dataset Overview
- Data Preparation
- Exploratory Data Analysis
- Statistical Analysis
- Business Findings
- Recommendations
- Conclusion

Introduction

Healthcare organizations generate vast amounts of patient, clinical, and billing data every day. Transforming this raw data into actionable insights through healthcare analytics is critical for modern medicine. It allows providers to transition from reactive care to proactive population health management, predict disease prevalence, optimize resource utilization, and ultimately improve the quality of patient care.

How data helps hospitals reduce costs?

Operating a hospital involves balancing massive clinical and financial complexities. Data analysis helps healthcare facilities identify inefficiencies and drive down costs without compromising patient outcomes by:

Optimizing Resource Allocation: Tracking admission types (Emergency vs. Elective) helps management predict staffing needs and bed availability, preventing expensive overtime labour or underutilized resources.

Identifying Cost Drivers: Analyzing billing amounts across different demographics and medical conditions allows hospitals to pinpoint why certain treatments or departments incur higher expenses.

Preventing Operational Bottlenecks: Data highlights patterns in length of stay and insurance utilization, enabling hospitals to streamline the discharge process and reduce administrative overhead.

Business Objective

The objective of this project is to analyze healthcare patient data to identify trends in demographics, disease prevalence, hospital admissions, insurance utilization, billing patterns, and treatment outcomes. Statistical techniques are used to validate whether observed differences are significant, enabling evidence-based healthcare decision-making.

The primary goal of this analysis is to:

Exploratory Data Analysis (EDA): To visually and numerically map out the demographic landscape of the patients and understand how billing varies across different groups.

Statistical Inference: To go beyond superficial data summaries and use rigorous statistical testing such as Independent Samples T-Tests and One-Way ANOVA to determine which variables such as admission types or gender actively drive healthcare costs or remain statistically irrelevant.

Dataset Overview

The dataset used in this project contains **10,000 patient healthcare records** representing various aspects of hospital operations, patient demographics, medical conditions, treatment details, insurance providers, and billing information. The data was analyzed to identify healthcare trends, evaluate financial patterns, and perform statistical testing to support data-driven decision-making.

The dataset provides a comprehensive view of patient admissions and healthcare costs, making it suitable for exploratory data analysis (EDA), descriptive statistics, and inferential statistical analysis.

1.Dataset Summary

Attribute	Details
Dataset Name	US Healthcare Dataset
Source	Kaggle
Total Records	10,000 Patients
Number of Features	15
File Format	Excel
Analysis Tool	Python (Jupyter Notebook)
Libraries Used	Pandas, NumPy, Matplotlib, Seaborn, SciPy

2. Dataset Variables

Variable	Description
Patient Name	Name of the patient
Age	Age of the patient (years)
Gender	Male or Female
Blood Type	Patient blood group
Medical Condition	Diagnosed disease or medical condition
Date of admission	Date of hospital admission
Doctor	Attending physician
Hospital	Hospital where treatment was provided
Insurance Provider	Insurance company covering treatment expenses
Billing Amount	Total treatment cost incurred by the patient
Room Number	Room in which patient was admitted
Admission Type	Emergency, Urgent, or Elective
Discharge Date	Date of patient discharge
Medication	Medication prescribed to the patient
Test Results	Medical test outcome (Normal, Abnormal, Inconclusive)

Data Preparation

Before performing the analysis, the dataset was carefully examined and cleaned to ensure accuracy and reliability.

The following preprocessing steps were performed:

- Imported the dataset into Python using the Pandas library.
- Examined the dataset structure, dimensions, and data types.
- Checked for missing values across all variables.
- Verified duplicate records and confirmed data consistency.
- Converted **Admission Date** and **Discharge Date** into datetime format.
- Created a new feature, **Length of Stay**, by calculating the difference between discharge and admission dates.
- Verified numerical and categorical variables before performing statistical analysis.

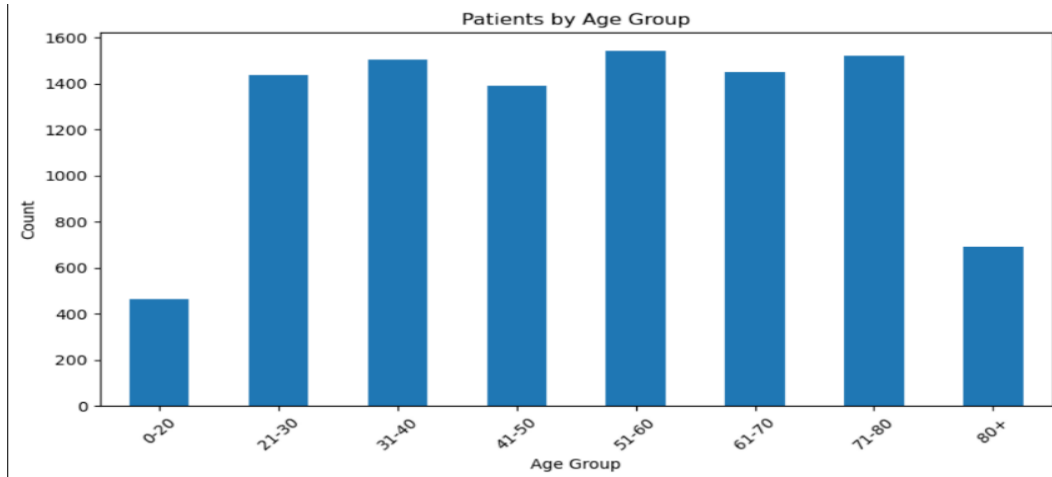
These preprocessing steps ensured that the dataset was clean, consistent, and suitable for exploratory analysis and hypothesis testing.

The analysis focused on answering the following business questions:

- What are the demographic characteristics of the patient population?
- Which medical conditions are most frequently diagnosed?
- Which insurance providers handle the highest number of patients?
- How do hospital billing amounts vary across different patient groups?
- Does patient gender significantly influence healthcare billing?
- Does admission type affect hospital billing costs?
- Is there an association between patient gender and hospital admission type?
- What actionable insights can support better hospital management and resource planning?

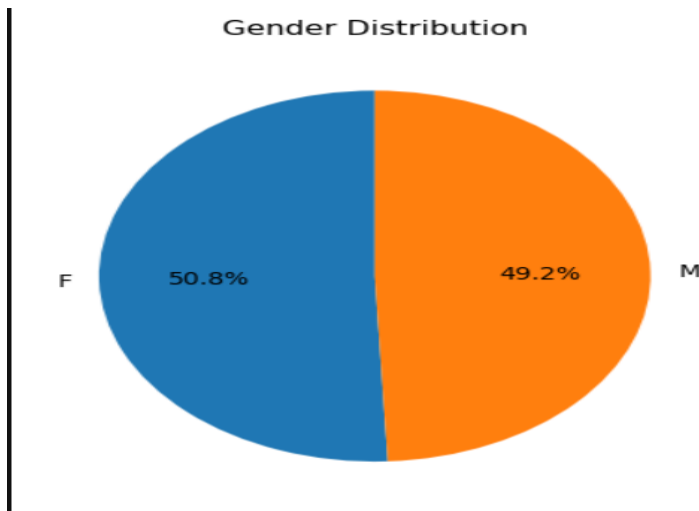
Exploratory Data Analysis

- **Age Distribution**



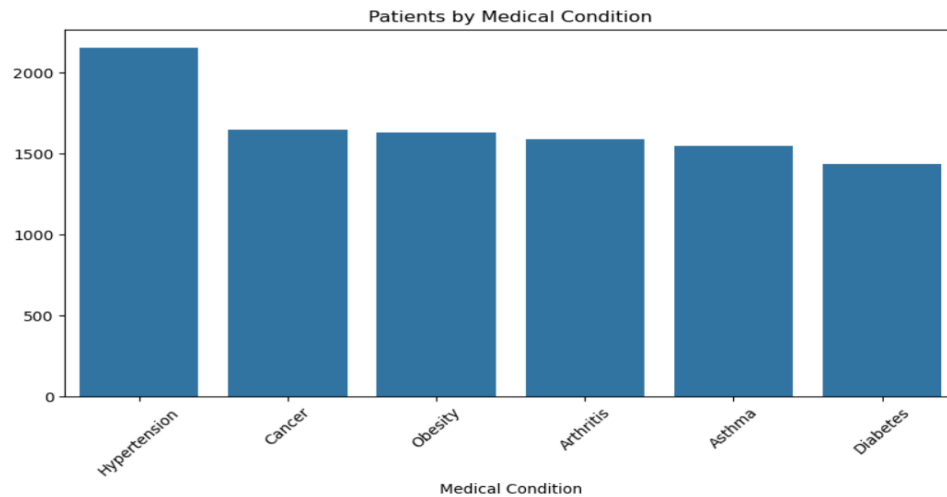
Insight: The most common patients are generally aged between 51-80

- **Gender Distribution**



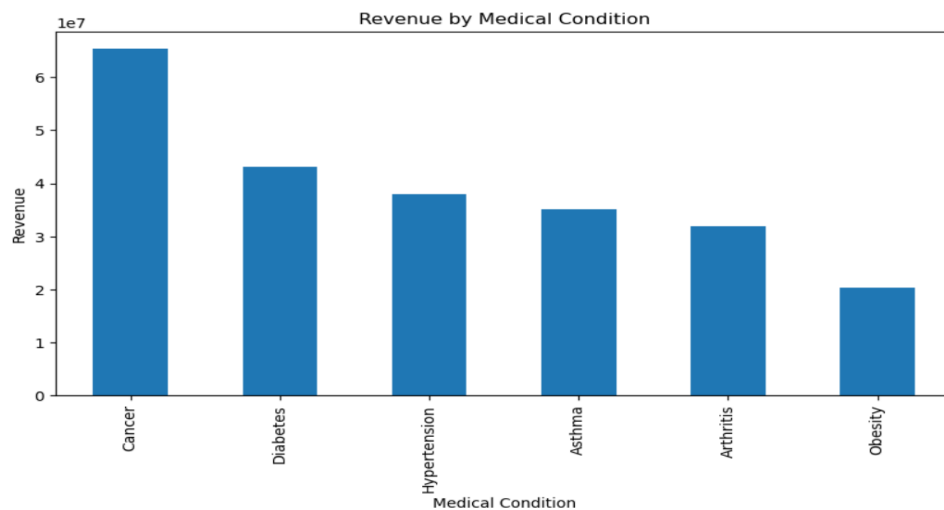
Insight: There is not that much difference in gender but males are slightly more than females

- **Medical Condition**



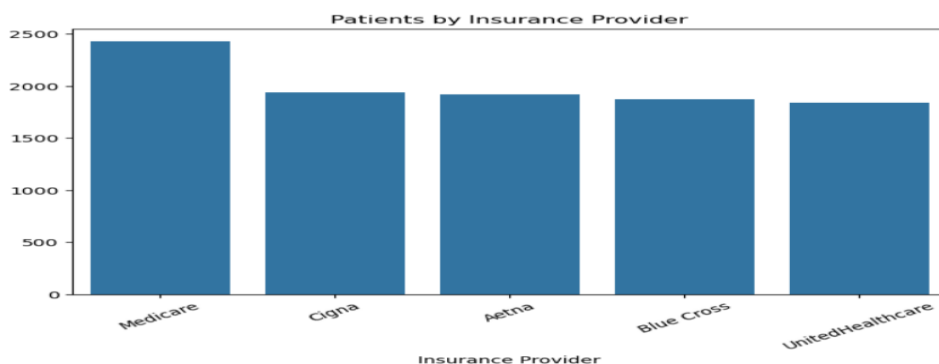
Insight: Hypertension is the most common disease.

- **Revenue by Medical Condition**



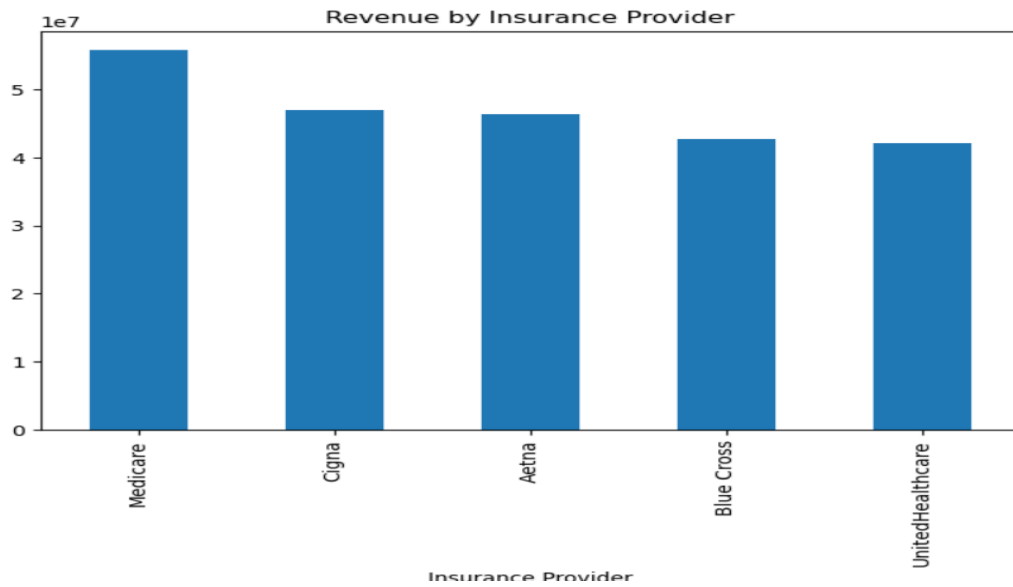
Insight: Cancer condition contributes the most billing revenue

- **Patients by Insurance Provider**

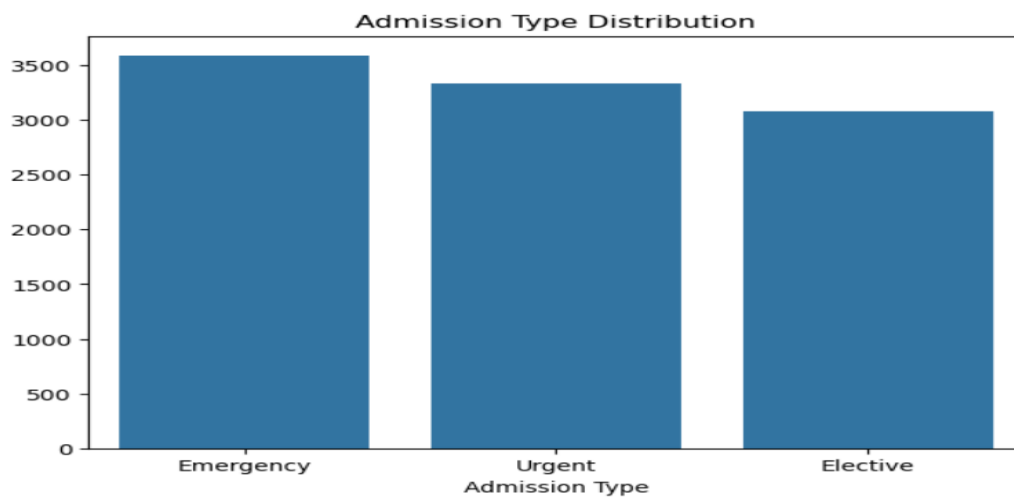


Insight: Medicare covers the highest number of patients.

- **Revenue by Insurance Provider**

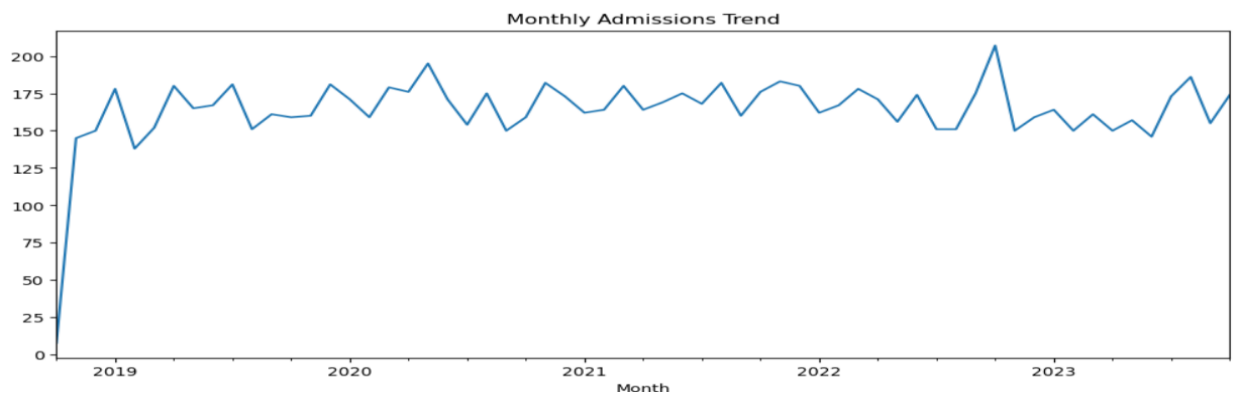


- **Admission Type Distribution**



Insight: Emergency admissions are highest

- **Monthly Admissions Trend**



Statistical Analysis

Test	Purpose	Result
Mean	Average Billing	✓
Median	Middle Value	✓
Standard Deviation	Spread	✓
Confidence Interval	Estimate Population Mean	✓
Independent t-test	Male vs Female Billing	✓
ANOVA	Admission Type vs Billing	✓
Chi-Square	Gender vs Admission Type	✓

- **Confidence Interval**

1. What is the likely range of the true average billing amount?

The average hospital billing amount was approximately **\$23,382**. Based on a 95% confidence interval, the true average billing amount is estimated to lie between **\$23,101 and \$23,662**, providing a reliable estimate of healthcare costs.

- **Independent Samples t-Test**

2. Does gender influence billing amount?

The independent samples t-test produced a **t-statistic of 0.3989** and a **p-value of 0.6900**. Since the p-value is greater than the significance level (0.05), there is insufficient statistical evidence to conclude that the average hospital billing amount differs significantly between male and female patients.

- **ANOVA Test**

3. Does Admission Type affect Billing?

The One-Way ANOVA test produced an **F-statistic of 8.6972** and a **p-value of 0.0002**. Since the p-value is less than the significance level (0.05), we reject the null hypothesis. This indicates that there is a statistically significant difference in the average hospital billing amount among at least one of the admission types (Emergency, Urgent, or Elective).

- **Chi-Square Test**

4. Is Gender associated with Admission Type?

The Chi-Square Test of Independence produced a **Chi-Square statistic of 13.5597** with **2 degrees of freedom** and a **p-value of 0.0011**. Since the p-value is less than the significance level (0.05), we reject the null hypothesis. This indicates a statistically significant association between patient gender and hospital admission type.

Key Business Findings

- Hypertension is the most common disease.
- Emergency admissions are highest.
- Medicare covers the largest number of patients.
- Gender does NOT significantly affect billing.
- Admission Type significantly affects billing.
- Gender and Admission Type are statistically associated.
- Billing amounts exhibited considerable variability, indicating diverse treatment costs across patients.

Recommendations

Recommendation 1

Increase hypertension awareness programs.

Recommendation 2

Allocate more resources to Emergency departments.

Recommendation 3

Review billing differences across admission types.

Recommendation 4

Monitor demographic admission trends.

Recommendation 5

Use statistical analysis for budgeting decisions.

Conclusion

- 10,000 Healthcare Records Analyzed

- 15+ Business Insights Generated
- 3 Hypothesis Tests Conducted
- 95% Confidence Interval Estimated
- 20+ Visualizations Created
- Actionable Business Recommendations Provided

This project demonstrates the application of Python, exploratory data analysis, and statistical inference to uncover meaningful healthcare insights. By combining data visualization with hypothesis testing, the analysis provides evidence-based findings that can support healthcare planning, operational improvements, and informed decision-making.